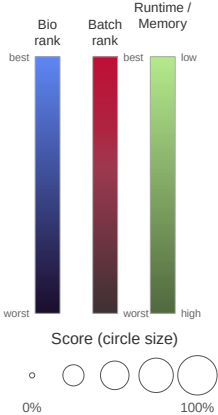


scRNA-seq integration benchmark -- Bio conservation & Batch correction ranking

Method	Language	Category	Output	Overall	Bio Conservation								Batch Correction					Scalability		
scGen	Python	Deep learning	Corrected embedding																	
scANVI	Python	Deep learning	Latent embedding																	
Seurat v5	R	Anchor-based (RPCA)	Corrected embedding																	
fastMNN	R	MNN-based correction	Corrected embedding																	
Scanorama	Python	Manifold alignment (MNN-based)	Corrected embedding																	
BBKNN	Python	Graph-based	Corrected graph																	
Harmony	R/Python	Linear correction (embedding)	Corrected PCA embedding																	
scVI	Python	Deep learning	Latent embedding																	
MNN	R/Python	MNN-based correction	Corrected matrix																	
ComBat	R/Python	Global linear model	Corrected matrix																	
LIGER	R/Python	Matrix factorization	Latent factors embedding																	



Overall rectangle (before bio): colour = bio Friedman rank; width = batch Friedman rank (narrow = best batch correction).

Bio/Batch dots: colour = per-metric rank; size = normalised score. Scalability dots: lighter/larger = faster or less RAM. Overall Scalability rect: colour = runtime rank (light=fast); width = memory rank (narrow=less RAM). batch_ASW & pc_regression inverted.